MAYO CLINIC FOR MEDICAL EDUCATION AND RESEARCH

Comprehensive Clinical Guide to High Blood Pressure and Patient Safety

Clinical Knowledge, Secondary Etiologies, Complications, and Rebound Warning

Official Reference: Mayo Clinic Clinical Disease Monograph (Arterial Hypertension)

Official Web Portal: https://www.mayoclinic.org/ar/diseases-conditions/high-blood-pressure

1. Primary (Essential) vs. Secondary Hypertension

A. Primary (Essential) Hypertension (90% to 95% of cases):

* Multifactorial etiology: Genetic inheritance, age-related arterial remodeling, high sodium, sedentary habi
* Develops insidiously over years and requires long-term chronic management.

B. Secondary Hypertension (5% to 10% of cases):

* Characterized by sudden onset, severe resistance to conventional drugs, and an identifiable underlying cau
* Can frequently be cured or substantially improved if the underlying pathology is addressed.

2. Key Causes of Secondary Hypertension

* Chronic kidney disease (CKD) and renal artery stenosis (Renovascular disease).
* Obstructive sleep apnea (OSA) causing nocturnal hypoxemia and sympathetic overdrive.
* Endocrine disorders: Cushing's syndrome, primary aldosteronism (Conn's syndrome), Pheochromocytoma.
* Medications & Substances: Oral contraceptives, NSAIDs (Ibuprofen), systemic corticosteroids, decongestants

3. Rebound Hypertension: Severe Danger of Abrupt Medication Cessation

* Ceasing antihypertensive medications abruptly triggers violent sympathetic rebound (Rebound Hypertension).
* Blood pressure can rapidly surge to levels significantly higher than baseline pre-treatment values.
* Fatal risks: Acute aortic dissection, hypertensive encephalopathy, acute myocardial infarction, fatal stro
* Mandatory Protocol: Any dosage taper or medication change must be executed strictly under medical supervis

4. Target Organ Damage (TOD) Pathology

* Heart: Left ventricular hypertrophy, ischemic coronary disease, congestive heart failure.
* Brain: Ischemic infarction, cerebral hemorrhage, lacunar infarcts, progressive vascular dementia.
* Kidneys: Arteriolar nephrosclerosis, glomerular sclerosis, proteinuria, irreversible ESRD.
* Eyes: Hypertensive retinopathy, optic disc edema (papilledema), retinal vein/artery occlusion.

5. Herbal Remedies & Complementary Medicine Facts

* Hibiscus (Karkadeh): Mild diuretic effect providing 2-7 mmHg reduction; hot and cold forms have equal effe
* Garlic (Allicin): Promotes minor nitric oxide production and mild vasodilation.
* CRITICAL MEDICAL DIRECTIVE: These substances are minor complementary dietary adjuncts ONLY.

* Under NO circumstances should herbal products replace prescribed pharmaceutical regimens.